1. Executive Summary

The analysis examined the data structure, quality, and global EV adoption trends. After minor preprocessing—removing an inconsistent percentage column and addressing year-related outliers—the data became clean and suitable for further analysis. The results indicate strong growth in EV adoption over time, with clear regional differences in deployment. Overall, the analysis provides a reliable foundation for forecasting and strategic EV market evaluation.

2. Introduction

- **Background and Context:** With rising climate concerns, electric vehicles (EVs) have become central to global sustainability efforts. This analysis examines regional and yearly EV adoption trends to understand how electric mobility has grown over time and across regions.
- **Purpose of the analysis:** This analysis examines regional and yearly electric vehicle adoption trends to identify key patterns and differences in global EV growth. It provides insights that support informed decision-making and future analytical or predictive modeling.
- **Scope of the study:** This study performs exploratory data analysis to examine regional and yearly EV adoption trends, focusing on data cleaning, summary statistics, and visualizations to understand growth patterns.

3. Problem Definition

- **Business Problem:** As electric vehicle adoption grows rapidly, stakeholders lack clear insights into regional growth rates and historical trends, making strategic planning and investment decisions challenging. This analysis addresses the problem by identifying key adoption patterns and regional differences to support data-driven decision-making.
- **Key Questions:**
    - How has EV adoption changed over the years?
    - Which regions show the highest and lowest levels of EV growth?
    - What insights can support strategic planning and EV market expansion?

4. Data Description

- **Data Source:** The data for this analysis was obtained from the International Energy Agency (IEA) Global EV.
- **Number of Records and variables:** The original dataset consists of 12,654 records and 9 variables. After removing outliers from the year variable, the dataset was reduced to 12,105 records and 8 variables.
- **Description of Key Features:**
    - **Region:** Geographic area of EV data for regional comparison.
    - **Year:** Observation year for trend analysis.
    - **Category:** Type of EV indicator (e.g., sales, stock).
    - **Mode:** Vehicle type or transport segment.
    - **Powertrain:** EV technology type (e.g., BEV, PHEV).
    - **Unit:** Measurement unit of the values.
    - **Value:** Numerical EV metric used for analysis.
    - **Parameter:** Specific EV metric being measured.

5. Data Preparation

- **Data Cleaning Steps:**
    - **Initial Inspection:** Reviewed structure, data types, and missing values using summary functions.
    - **Missing Values Check:** Verified completeness across all variables.
    - **Removed percentage Column:** Dropped due to inconsistency.
    - **Outlier Detection:** Used boxplot to identify year outliers.
    - **Outlier Removal:** Filtered invalid year values.
    - **Post-Cleaning Check:** Revalidated shape and summary statistics.
- **Outliers Handling:**
    - **Boxplot Detection:** Used boxplot to identify year outliers.
    - **Irregular Values Identified:** Detected years outside the valid range.
    - **Outliers Removed:** Filtered abnormal year records.
    - **Revalidation:** Rechecked dataset shape and summary statistics.
- **Data transformations:** The value column was originally stored as an object data type and was later converted to an integer data type for accurate numerical analysis.

6. Exploratory Data Analysis

6.1 Univariate Analysis

- **Year:** Analyzed distribution over time and checked for outliers.
- **Region:** Used value counts to assess record distribution by geography.
- **Value:** Summarized with descriptive statistics to assess central tendency and spread.

6.2 Bivariate Analysis

- **Year & Value:** Aggregated EV stock and sales by year to assess trends; visualized using a **line plot**.
- **Region & Value:** Compared total EV values across regions (excluding "World"); visualized using a **log-scale bar plot**.
- **Mode & Value:** Analyzed EV distribution by transport mode; visualized using a **pie chart**.

6. 3 Multivariate Analysis

Multivariate analysis was not performed in the analysis, as the analysis focused primarily on univariate and bivariate exploration.

7. Key Insights

- **Strong Upward Trend:** EV stock and sales show consistent growth over time, with rapid acceleration after 2015.
- **Positive Time Relationship:** There is a clear positive relationship between year and EV value, confirming steady global expansion of electric mobility.
- **Regional Disparities:** EV adoption is concentrated in a few key regions, while others contribute relatively smaller shares.
- **Mode Dominance & Skewness:** Passenger cars dominate EV deployment, and the distribution of values is highly skewed toward recent years and leading regions.

8. Recommendations

- **Expand Charging Infrastructure:** Rapid EV growth requires stronger charging networks and grid capacity to support increasing demand.
- **Target Underperforming Regions:** Since adoption is concentrated in a few regions, policymakers should introduce focused incentives and support programs to promote balanced regional growth.
- **Accelerate Commercial Vehicle Electrification:** As passenger cars dominate EV adoption, additional measures are needed to encourage electrification of buses and trucks.
- **Adopt Data-Driven Planning:** Continued monitoring of EV trends is essential for informed policy decisions, investment planning, and sustainable market expansion.

9. Conclusion

The analysis confirms strong and accelerating growth in global electric vehicle (EV) adoption, with significant increases in stock and sales over time, especially in recent years. Growth is concentrated in a few leading regions, and passenger cars dominate overall EV deployment. Overall, the findings indicate a steady global shift toward electric mobility and provide a solid foundation for future forecasting and strategic decision making.

10. Weakness in the Project

- **No Policy or Economic Context Integration:** The analysis does not link trends to external factors such as policy changes, economic conditions, or Infrastructure development.